Class 08: Breast Cancer Mini Project

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Table of contents

Background

The goal of this mini-project is for you to explore a complete analysis using the unsupervised learning techniques covered in class.

Today we will analyze a data-set from fine needle aspiration (FNA) of a breast mass.

Data Import

The data is made available as a CSV file, read this in using `read.csv()`.

```r
fna.data <- "WisconsinCancer.csv"
 wisc.df <- read.csv(fna.data, row.names=1)
```

```r
head(wisc.df, 3)
```

```
         diagnosis radius_mean texture_mean perimeter_mean area_mean
842302           M       17.99        10.38          122.8      1001
842517           M       20.57        17.77          132.9      1326
84300903         M       19.69        21.25          130.0      1203
```

```
         smoothness_mean compactness_mean concavity_mean concave.points_mean
842302           0.11840          0.27760         0.3001             0.14710
842517           0.08474          0.07864         0.0869             0.07017
84300903         0.10960          0.15990         0.1974             0.12790
         symmetry_mean fractal_dimension_mean radius_se texture_se perimeter_se
842302          0.2419                0.07871    1.0950     0.9053        8.589
842517          0.1812                0.05667    0.5435     0.7339        3.398
84300903        0.2069                0.05999    0.7456     0.7869        4.585
         area_se smoothness_se compactness_se concavity_se concave.points_se
842302    153.40      0.006399        0.04904      0.05373           0.01587
842517     74.08      0.005225        0.01308      0.01860           0.01340
84300903   94.03      0.006150        0.04006      0.03832           0.02058
         symmetry_se fractal_dimension_se radius_worst texture_worst
842302       0.03003             0.006193        25.38         17.33
842517       0.01389             0.003532        24.99         23.41
84300903     0.02250             0.004571        23.57         25.53
         perimeter_worst area_worst smoothness_worst compactness_worst
842302             184.6       2019           0.1622            0.6656
842517             158.8       1956           0.1238            0.1866
84300903           152.5       1709           0.1444            0.4245
         concavity_worst concave.points_worst symmetry_worst
842302            0.7119               0.2654         0.4601
842517            0.2416               0.1860         0.2750
84300903          0.4504               0.2430         0.3613
         fractal_dimension_worst
842302                   0.11890
842517                   0.08902
84300903                 0.08758
```

Make sure we remove or exclude the diagnosis column from the data-set that we use for further analysis - this is the expert diagnosis as either M or B:

```
wisc.data <- wisc.df[,-1]
diagnosis <- as.factor(wisc.df$diagnosis)
```

Exploratory Data Analysis

> Q1. How many observations are in this dataset?

569 observations.

```r
nrow(wisc.data)
```

```
[1] 569
```

> Q2. How many of the observations have a malignant diagnosis?

212 malignant observations.

```r
table(wisc.df$diagnosis)
```

```
  B   M 
357 212 
```

```r
sum(wisc.df$diagnosis == "M")
```

```
[1] 212
```

> Q3. How many variables/features in the data are suffixed with `_mean`?

10 features are suffixed with _mean (see below).

We can use the `grep()` function to help us here:

```r
colnames(wisc.data)
```

```
 [1] "radius_mean"             "texture_mean"           
 [3] "perimeter_mean"          "area_mean"              
 [5] "smoothness_mean"         "compactness_mean"       
 [7] "concavity_mean"          "concave.points_mean"    
 [9] "symmetry_mean"           "fractal_dimension_mean" 
[11] "radius_se"               "texture_se"             
[13] "perimeter_se"            "area_se"                
[15] "smoothness_se"           "compactness_se"         
[17] "concavity_se"            "concave.points_se"      
[19] "symmetry_se"             "fractal_dimension_se"   
[21] "radius_worst"            "texture_worst"          
[23] "perimeter_worst"         "area_worst"             
[25] "smoothness_worst"        "compactness_worst"      
[27] "concavity_worst"         "concave.points_worst"   
[29] "symmetry_worst"          "fractal_dimension_worst"
```

```
length(grep("_mean", colnames(wisc.data), value = TRUE))
```

```
[1] 10
```

Principal Component Analysis (PCA)

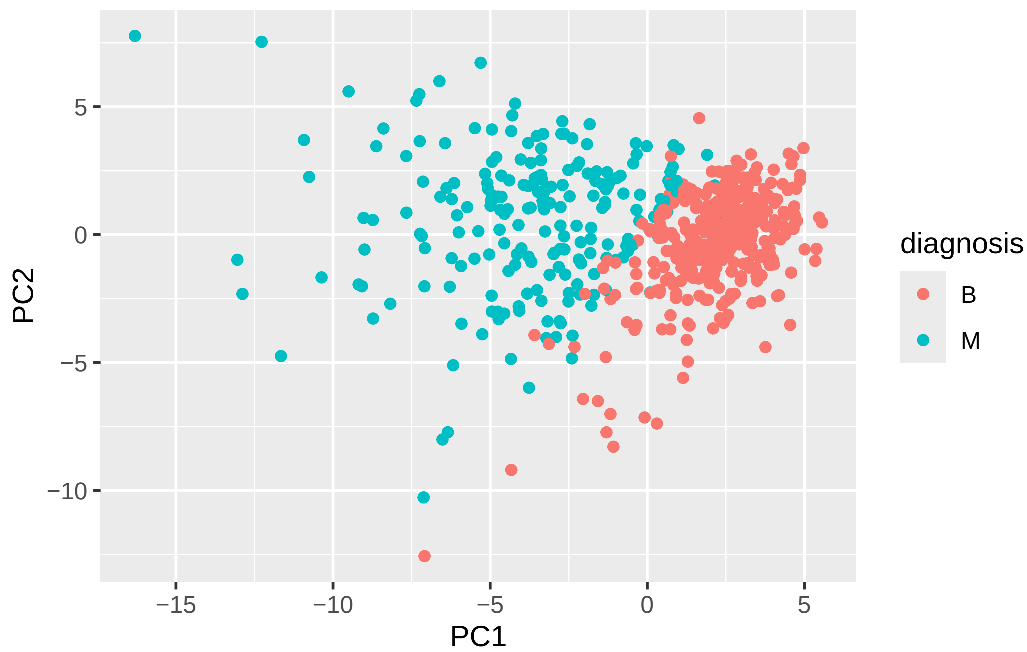
We need to scale our data before PCA with the `scale=TRUE` argument to `prcomp()`.

```
wisc.pr <- prcomp(wisc.data, scale=TRUE)
summary(wisc.pr)
```

```
Importance of components:
                          PC1    PC2     PC3     PC4     PC5     PC6     PC7
Standard deviation     3.6444 2.3857 1.67867 1.40735 1.28403 1.09880 0.82172
Proportion of Variance 0.4427 0.1897 0.09393 0.06602 0.05496 0.04025 0.02251
Cumulative Proportion  0.4427 0.6324 0.72636 0.79239 0.84734 0.88759 0.91010
                           PC8    PC9    PC10   PC11    PC12    PC13    PC14
Standard deviation     0.69037 0.6457 0.59219 0.5421 0.51104 0.49128 0.39624
Proportion of Variance 0.01589 0.0139 0.01169 0.0098 0.00871 0.00805 0.00523
Cumulative Proportion  0.92598 0.9399 0.95157 0.9614 0.97007 0.97812 0.98335
                          PC15    PC16    PC17    PC18    PC19    PC20   PC21
Standard deviation     0.30681 0.28260 0.24372 0.22939 0.22244 0.17652 0.1731
Proportion of Variance 0.00314 0.00266 0.00198 0.00175 0.00165 0.00104 0.0010
Cumulative Proportion  0.98649 0.98915 0.99113 0.99288 0.99453 0.99557 0.9966
                          PC22    PC23   PC24    PC25    PC26    PC27    PC28
Standard deviation     0.16565 0.15602 0.1344 0.12442 0.09043 0.08307 0.03987
Proportion of Variance 0.00091 0.00081 0.0006 0.00052 0.00027 0.00023 0.00005
Cumulative Proportion  0.99749 0.99830 0.9989 0.99942 0.99969 0.99992 0.99997
                          PC29    PC30
Standard deviation     0.02736 0.01153
Proportion of Variance 0.00002 0.00000
Cumulative Proportion  1.00000 1.00000
```

Let's see the "PC score plot"

```
library(ggplot2)
ggplot(wisc.pr$x) +
  aes(PC1, PC2, col=diagnosis) +
  geom_point()
```



> Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

0.4427 –> 44.27% of the original variance is captured by PC1.

> Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

We need 3 principle components (3 PCs) to reach at least 70%. Cumulative Proportion of PC3 = 0.72636 = 72.64%.

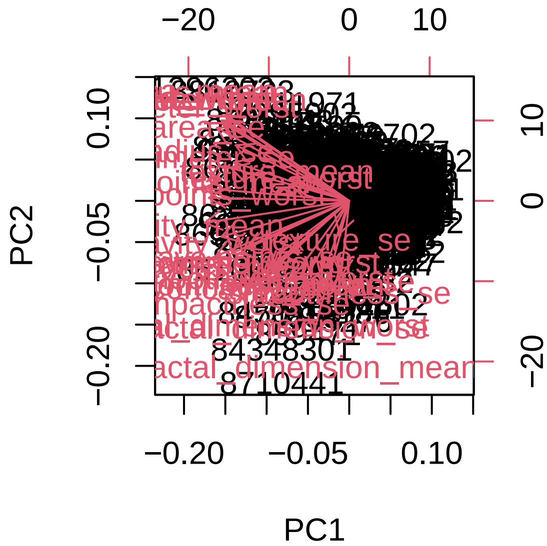
> Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

We need 7 PCs to describe at least 90% of the original variance in the data. PC7 = 0.91010 = 91.01%.

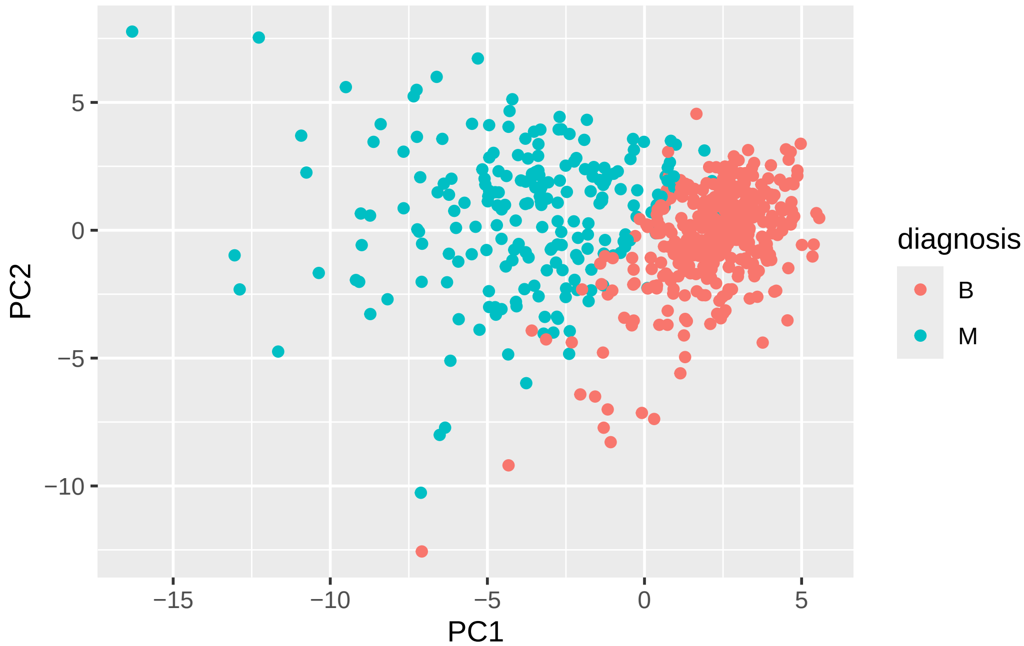
> Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

The plot is difficult to understand because the overlapping labels and dense clustering make it visually cluttered and hard to interpret.

```r
biplot(wisc.pr)
```



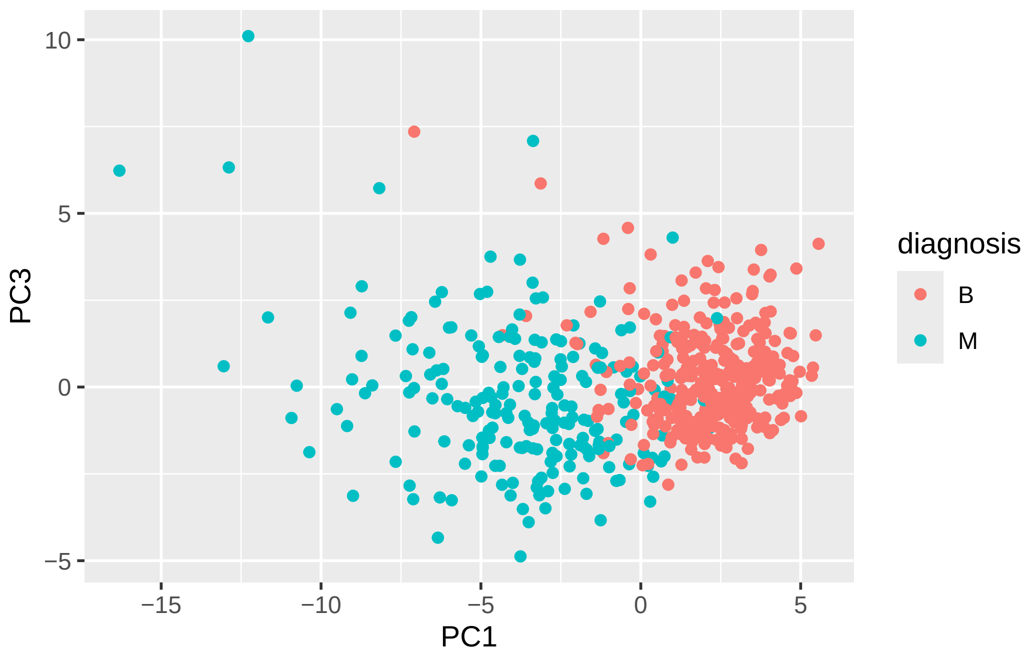

```r
library(ggplot2)
ggplot(wisc.pr$x) +
  aes(PC1, PC2, col=diagnosis) +
  geom_point()
```



> Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

The PC1 vs PC3 plot shows a similar separation between benign and malignant group. The PC1 vs PC2 plot shows the separation is less distinct, indicating PC2 better displays the variation that distinguishes the two classes.

```
ggplot(wisc.pr$x) +
  aes(PC1, PC3, col=diagnosis) +
  geom_point()
```



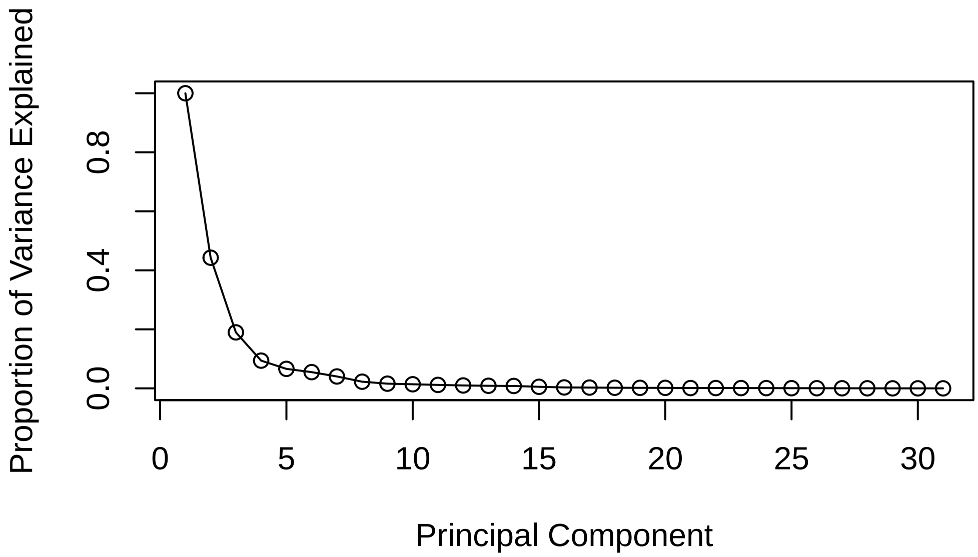

```r
pr.var <- wisc.pr$sdev^2
  head(pr.var)
```

```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

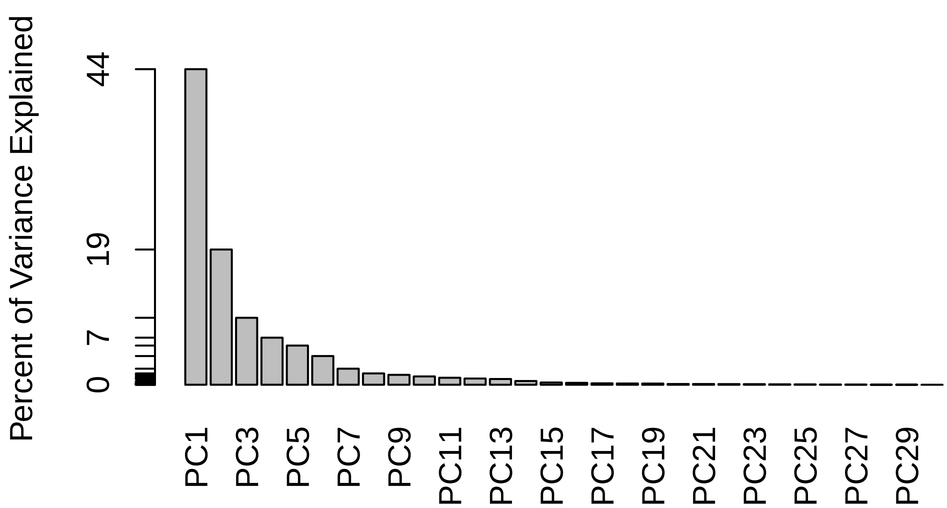
```r
pve <- pr.var / sum(pr.var)
pve
```

```
 [1] 4.427203e-01 1.897118e-01 9.393163e-02 6.602135e-02 5.495768e-02
 [6] 4.024522e-02 2.250734e-02 1.588724e-02 1.389649e-02 1.168978e-02
[11] 9.797190e-03 8.705379e-03 8.045250e-03 5.233657e-03 3.137832e-03
[16] 2.662093e-03 1.979968e-03 1.753959e-03 1.649253e-03 1.038647e-03
[21] 9.990965e-04 9.146468e-04 8.113613e-04 6.018336e-04 5.160424e-04
[26] 2.725880e-04 2.300155e-04 5.297793e-05 2.496010e-05 4.434827e-06
```

```r
plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```




```
barplot(pve, ylab = "Percent of Variance Explained",
     names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
axis(2, at=pve, labels=round(pve,2)*100 )
```

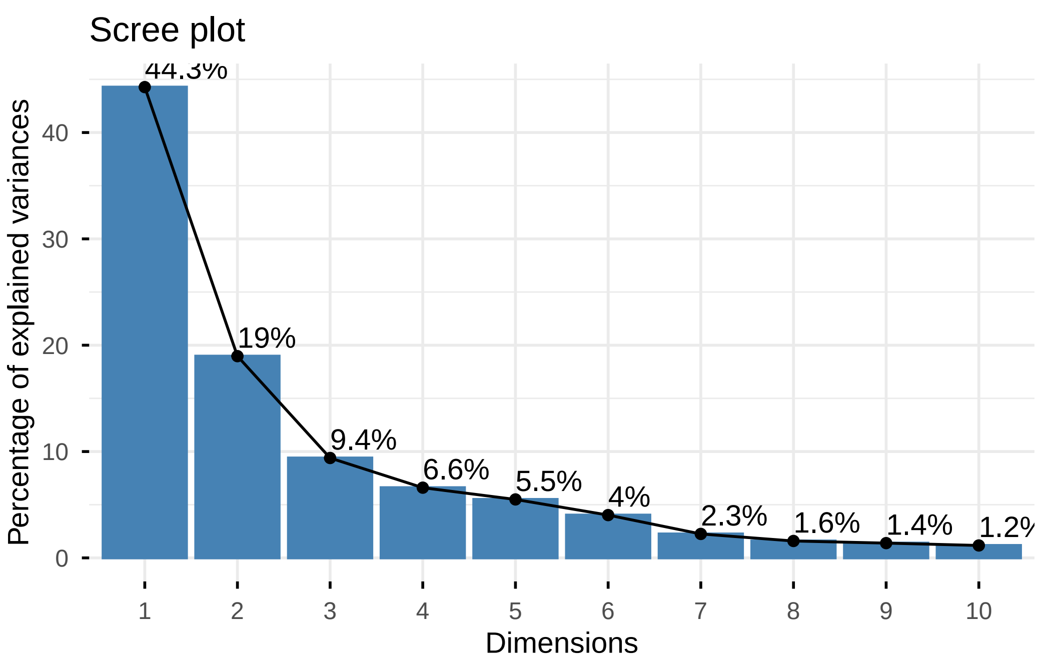


```r
library(factoextra)
```

```
Welcome to factoextra!

Want to learn more? See two factoextra-related books at https://www.datanovia.com/en/product/
guide-to-principal-component-methods-in-r/
```

```r
fviz_eig(wisc.pr, addlabels = TRUE)
```



> Q9. For the first principal component, what is the component of the loading vector (i.e. wisc.pr$rotation[,1]) for the feature concave.points_mean? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

The loading for concave.points_mean on PC1 is one of the largest, and no other features have significantly larger contributions. This feature is one of the strongest contributors to the first principal component.

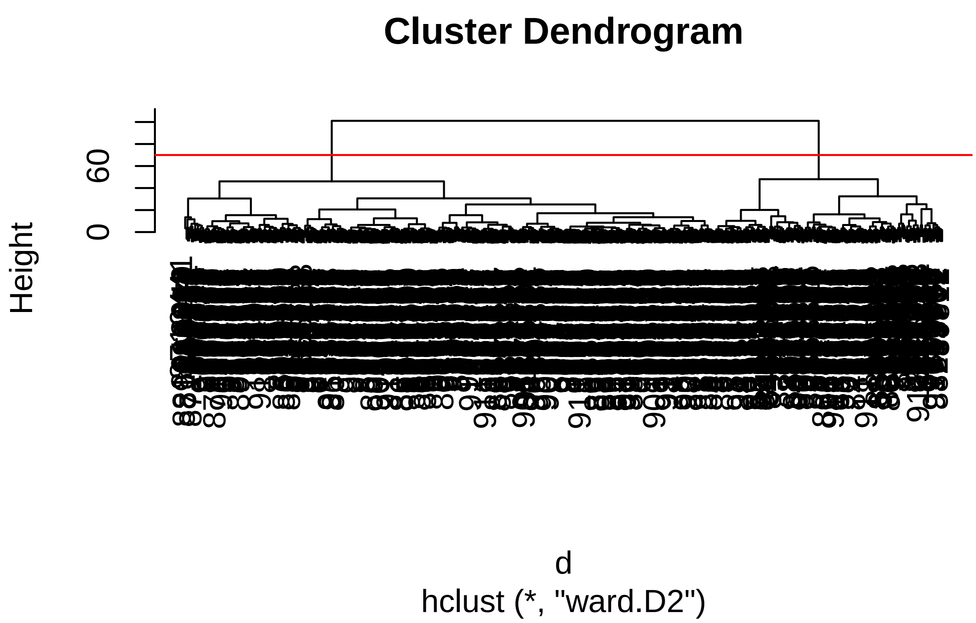
Hierarchial Clustering

```
data.scaled <- scale(wisc.data)
```

```
data.dist <- dist(data.scaled)
```

```
wisc.hclust <- hclust(data.dist, method = "complete")
```

```
d <- dist(wisc.pr$x[,1:4])
wisc.pr.hclust <- hclust(d, method="ward.D2")
plot (wisc.pr.hclust)
abline(h=70, col="red")
```

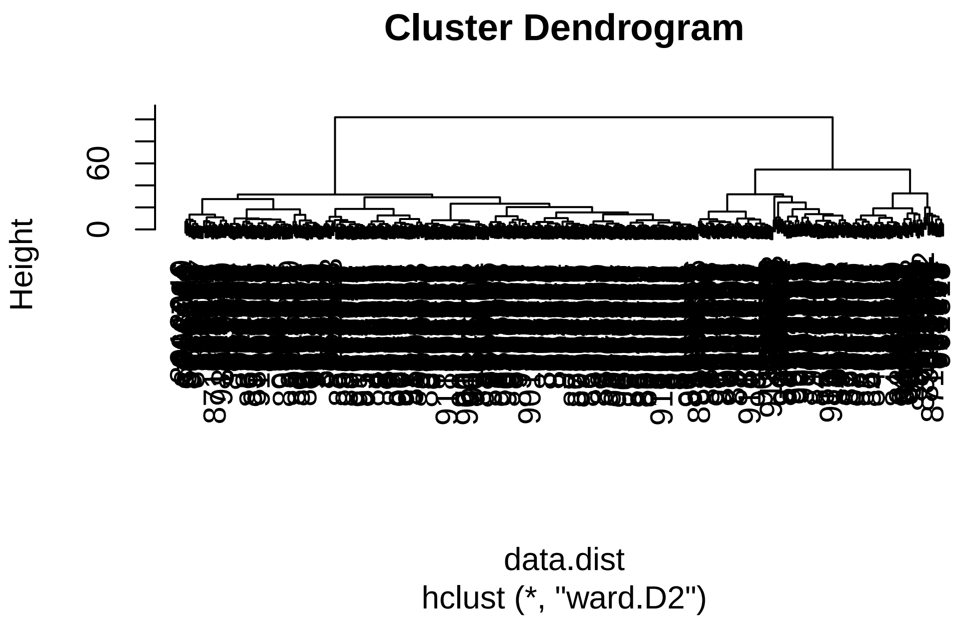


> Q10: Using the plot() and abline() functions, what is the height at which the clustering model has 4 clusters?

Height = 70, which is where the dendrogram is cut to yield 4 clusters.

```
wisc.hclust.single <- hclust(data.dist, method = "single")
wisc.hclust.complete <- hclust(data.dist, method = "complete")
wisc.hclust.average <- hclust(data.dist, method = "average")
wisc.hclust.ward <- hclust(data.dist, method = "ward.D2")

plot(wisc.hclust.ward)
```



> Q12: Which method gives your favorite results for the same data.dist dataset? Explain your reasoning.

The "ward.D2" method gives the best results because it produces balanced and clearly separated clusters by minimizing variance within the cluster, making the grouping easier to interpret.

Combining Methods

```r
grps <- cutree(wisc.pr.hclust, h=70)
table(grps)
```

```
grps
  1   2
171 398
```

How does this clustering `grps` correspond to the expert `diagnosis`?

```r
table(diagnosis)
```

```
diagnosis
  B   M 
357 212 
```

```r
table(diagnosis, grps)
```

```
         grps
diagnosis   1   2
        B   6 351
        M 165  47
```

```r
wisc.pr.hclust <- hclust(dist(wisc.pr$x[, 1:7]), method = "ward.D2")
```

```r
wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, k = 2)
```

```r
table(wisc.pr.hclust.clusters)
```

```
wisc.pr.hclust.clusters
  1   2 
216 353 
```

```r
table(wisc.pr.hclust.clusters, diagnosis)
```

```
                       diagnosis
wisc.pr.hclust.clusters   B   M
                      1  28 188
                      2 329  24
```

> Q13: How well does the newly created hclust model with two clusters separate out the two "M" and "B" diagnoses?

The PCA based hierarchical clustering separates the data well, with most malignant samples grouped in one cluster and most benign samples in the other, though there are some misclassifications.

```r
wisc.hclust.clusters <- cutree(wisc.hclust, k = 4)
table(wisc.hclust.clusters, diagnosis)
```

```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

> Q14: How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the table() function to compare the output of each model (wisc.hclust.clusters and wisc.pr.hclust.clusters) with the vector containing the actual diagnoses.

The clustering without PCA performs worse, showing more mixing between benign and malignant samples, whereas PCA based clustering provides clearer separation between the two groups.

Prediction

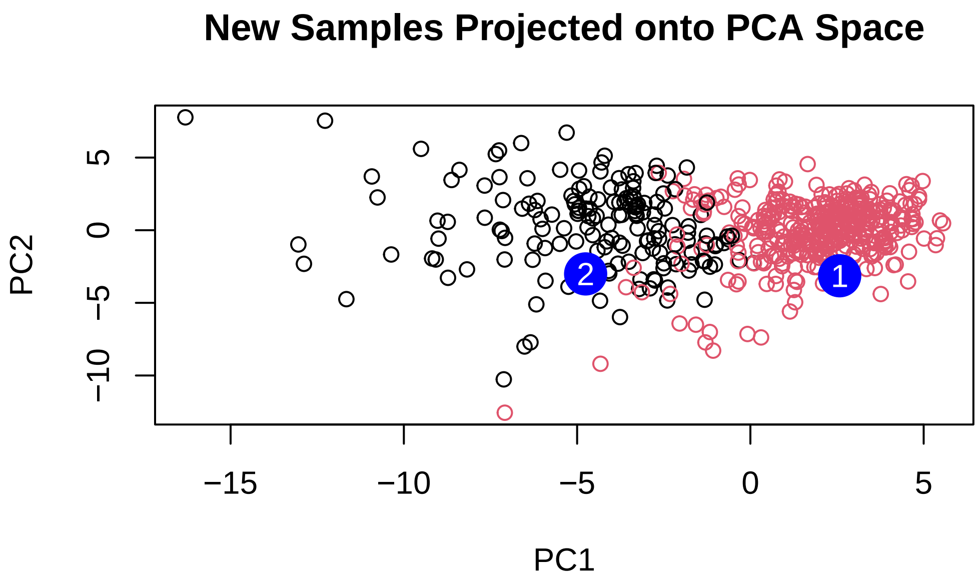
```r
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)

npc <- predict(wisc.pr, newdata = new)

plot(wisc.pr$x[, 1:2], col = grps,
     xlab = "PC1", ylab = "PC2",
     main = "New Samples Projected onto PCA Space")

points(npc[, 1], npc[, 2], col = "blue", pch = 16, cex = 3)

text(npc[, 1], npc[, 2], labels = c(1, 2), col = "white")
```



Q16: Which of these new patients should we prioritize for follow up based on your results?

Patient 1 should be prioritized for follow up because it falls closer to the malignant associated cluster in the PCA plot.